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SUPERSEDING
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MILITARY SPECIFICATION

SALINOMETER EQUIPMENT,

EVAPORATOR, SHIPBOARD USE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force

1 SCOPE

1 1 This specification covers test equipment for use in measuring the density of the brine discharged from a sea water distilling unit.

2 APPLICABLE DOCUMENTS

2 1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein·

SPECIFICATIONS

FEDERAL

- PPP-B-585 - Boxes, Wood, Wirebound
- PPP-B-591 - Boxes, Fiberboard, Wood-Cleated
- PPP-B-601 - Boxes, Wood, Cleated-Plywood
- PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
- PPP-B-636 - Box, Fiberboard
- PPP-B-676 - Boxes, Set-Up, Paperboard
- PPP-T-60 - Tape, Pressure-Sensitive Adhesive, Waterproof, for packaging and sealing.
- PPP-T-97 - Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

- MIL-P-116 - Preservation, Methods of
- MIL-B-10377 - Boxes: Wood-Cleated, Veneer, Paper Overlaid
- MIL-L-10547 - Liners, Case, Waterproof

STANDARDS

MILITARY

- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
- MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

2 2 Other publications. - The following documents form a part of this specification to the extent specified herein Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply

AMERICAN SOCIETY FOR TESTING MATERIALS

- B 122 - Copper-Nickel-Zinc Alloy (Nickel Silver) and Copper-Nickel Alloy Plate, Sheet, Strip, and Rolled Bar
- B 127 - Nickel-Copper Alloy Plate, Sheet, and Strip
- B 141 - Electrodeposited Coatings of Nickel and Chromium on Copper and Copper-Base Alloys
- B 149 - Leaded Nickel Brass (Leaded Nickel Silver) and Leaded Nickel Bronze (Leaded Nickel Silver) Sand Castings
- B 151 - Copper-Nickel-Zinc Alloy (Nickel Silver) Rod and Bar
- B 164 - Nickel-Copper Alloy Rods and Bars

(Application for copies should be addressed to the Superintendent of Documents Government Printing Office, Washington 25, D C)

OFFICIAL CLASSIFICATION COMMITTEE

- Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St , New York 16, N Y)

3. REQUIREMENTS

3 1 Material - Materials shall be as specified herein Materials not definitely specifed shall be of the best quality and suitable for the purpose intended

FSC 6630

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3.2 Component parts. - The salinometer test equipment shall consist of the following items which may be procured individually as required (see 6.1):

(a) Four hydrometers.
(b) One thermometer.
(c) One test pot.

3.2.1 Hydrometers. - Each hydrometer shall be not more than 12 inches nor less than 10 inches in length and not more than 1-3/4 inches at the largest diameter. The stem above the hydrometer float shall be square measuring not less than 1/4 inch on a side. The hydrometer shall be made from commercial brass; leaded nickel brass or bronze in accordance with Publication ASTM B 149; copper-nickel-zinc alloy in accordance with Publication ASTM B 122 or Publication ASTM B 151; or nickel-copper alloy in accordance with Publication ASTM B 127 or Publication ASTM B 164. Brass if used shall be nickel plated in accordance with Publication ASTM B 141.

3.2.1.1 Scales - Each hydrometer shall carry four density scales, one on each face of the square stem. Each scale shall be marked to indicate the density of the brine in thirty-seconds from 0 to at least 3 but not more than 5. The distance between major divisions shall be at least 3/4 inch and shall be subdivided by three minor division marks. Scale markings shall be engraved or stamped on the stem Each major scale marking shall be numbered. Numbers shall be as large as practicable.

3 2.1.2 The hydrometers shall have their respective scales located to indicate the density of the brine at the following temperatures:

(a) Hydrometer 1 - 110°F., 115°F., 120°F. and 125°F.
(b) Hydrometer 2 - 125°F., 130°F., 135°F and 140°F.
(c) Hydrometer 3 - 140°F., 145°F., 150°F. and 155°F.
(d) Hydrometer 4 - 155°F . 160°F., 165°F. and 170°F.

This temperature shall be clearly marked at the top of the hydrometer stem on the same face as the corresponding scale.

3.2.1.3 One thirty-second on the hydrometer scale shall represent a brine density equivalent to the density obtained by dissolving one part by weight certified pure NaCl (ACS Grade) in 31 parts by weight distilled water. The hydrometer shall therefore be calibrated at the various temperatures using multiples of the above mixture.

3 2 2 Thermometer. - The thermometer assembly shall consist of a self-indicating liquid-in-glass thermometer tube and bulb mounted on a backing strip. A clip shall be fastened to the backing strip. This clip shall be designed to clamp the thermometer assembly in a vertical postion in the test pot. The backing strip shall extend beyond the ends of the glass tube and shall have engraved or stamped on it the thermometer scale A guard shall be fastened to the lower part of the backing strip so as to protect the thermometer tube and bulb below the lowest scale reading. The complete thermometer assembly shall measure between 8 and 10 inches in length and not over 1-1/4 inches in width

3.2.2.1 Glass tube and bulb. - The thermometer tube and bulb shall have been adequately annealed. The tube shall be of a magnifying-lens type with a white enamel strip at the back. The width of the tube shall be not less than 3/16 inch. The magnifying column shall have a reading angle of not less than 5 degrees in a plane at right angles to the axis of the glass tube. The glass tube shall be centrally located with respect to the backing strip and a means shall be provided to prevent axial displacement of the tube. The tube shall be of sufficient length so that the distance between the bulb and the lowest scale graduation is at least 2 inches.

3.2.2.2 Indicating medium - Mercury shall not be used The indicating liquid shall appear as a red column. The liquid shall be nontoxic and shall not require special protective devices either from inhaling fumes or when handling. No separation of liquid and coloring shall be evident as a result of continuous exposure to temperatures within the scale range. The liquid shall have a boiling point of not less than 50°F. above the maximum scale graduation. Material used for indicating liquid shall be specifically approved by the bureau or agency concerned. Each thermometer shall be furnished with instructions describing the recommended procedure for rejoining a broken indicating column.

3 2.2.3 Scale. - The scale shall be not less than 3-3/4 inches long. The scale shall be neatly and permanently marked with black graduations and figures on a dull silvered, or dull white backing strip.

3 2 2.4 Graduations and markings - The bore of the thermometer shall bisect the graduation lines with every 10 degree line slightly longer than the intermediate lines. The numerals shall be centrally located in relation to the lines. The range of the thermometer shall be between at least 100°F. and 220°F. There shall be 2 degrees between graduation lines and 10 degrees between figures The distance between 2 degree graduations shall be not less than 1/16 inch.

3 2.2.5 Backing strip. - The backing strip shall be not less than 0.035 inch in thickness, shall be of sufficient length to extend beyond the ends of the

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thermometer tube and bulb at least 1/2 inch on each end and shall be not more than 1-1/4 inches in width. The top of the backing strip shall contain a hole at least 1/4 inch in diameter which may be used to hang the thermometer when not in use.

3 2 2.6 Guard. - A guard shall be provided to protect that portion of the thermometer tube and bulb which will be immersed in the brine solution being tested. This guard shall not prevent direct contact of the brine and the thermometer bulb. The guard shall extend from the lowest scale reading to the bottom of the backing strip

3 2 2.7 Spring clip. - A spring clip shall be provided to hold the thermometer in a vertical position in the brine pot. The clip shall locate the thermometer in the brine pot so that the lowest scale graduation will be above the lip of the pot and the thermometer bulb will be at least 1-3/4 inches below the lip of the pot.

3 2.2.8 Materials for thermometer assembly. - Except for the glass tube and bulb the thermometer assembly including the backing strip, guard and spring clip shall be made of commercial brass Brass shall be nickel plated in accordance with Publication ASTM B 141.

3 2 3 Test Pot - The test pot shall be approximately 12-1/2 inches above the base and shall have a maximum internal diameter of 2-5/8 inches. The pot shall have a wooden handle attached to the side. The handle shall be designed to permit carrying the pot, full of hot water, without discomfort to the bare hand. A base approximately 2 inches larger in diameter than the pot shall be attached to the bottom of the pot. A slide shall be fastened inside the pot This slide shall be designed and located to hold the thermometer, utilizing the thermometer spring clip, so that the lowest thermometer scale graduation will be above the lip of the pot

3.2 3.1 Material for test pot. - The test pot, slide and base shall be made from commercial brass not less than 0 035 inch thick.

3 3 Performance. -

3 3 1 Accuracy - The thermometer shall indicate the correct temperature within ± 1 of the smallest scale divisions when tested as specified in 4.6 1.1.

3.3 1.1 The hydrometer shall indicate the correct density within ± 1/4 of the smallest scale division throughout the entire range of the hydrometer.

3.3 1.2 Each hydrometer shall withstand submergence in 212°F water for an indefinite period without distortion of any kind.

3.3.2 Time lag. - The time lag of the thermometer shall not exceed 10 seconds.

3.4 Workmanship - Workmanship shall be first class in every respect.

4 QUALITY ASSURANCE PROVISIONS

4 1 The supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specifed, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4 2 Lot - For purposes of sampling, a lot shall consist of all equipment produced at one time.

4 3 Sampling for acceptance inspection -

4 3 1 Sampling for group A examination and tests. - Sample equipment shall be selected from each lot in accordance with Standard MIL-STD-105, inspection level II, AQL equal 1 5 percent The examination and tests shall be as specified in 4.4 and 4 5 1.

4 3 2 Sampling for group B tests - Sample equipment shall be selected from each lot in accordance with Standard MIL-STD-105, inspection level L-8, AQL equal 4 0 percent The tests shall be as specified in 4 5 2.

4.4 Examination. - Equipment selected in accordance with 4.3.1 shall be examined to determine conformance to the requirements of this specification not involving tests.

4.5 Group A and B tests. -

4.5.1 Group A examination and tests. - Equipment selected in accordance with 4.3.1 shall be subjected to each of the following group A tests·

Group A tests	Reference
Examination	4.4
Accuracy, thermometer	4.6.1.1
Accuracy, hydrometer	4.6.1.2

4.5.2 Group B tests. - Equipment selected in accordance with 4.3.2 shall be subjected to each of the following group B tests:

Group B tests	Reference
Accuracy - Thermometer	4.6 1.1
Accuracy - hydrometer	4.6.1.2

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Group B tests	Reference
Hydrometer exposure to water at 212°F	4.6.1.3
Thermometer time lag	4.6.2

4.6 Tests. -

4.6.1 Accuracy -

4.6.1.1 Thermometers - Thermometers shall be tested at 100, 150 and 200°F. to determine conformance with 3.3.1.

4.6.1.2 Hydrometers - Hydrometers shall be tested at 1, 2 and 3 thirty-seconds in the solution specified in 3 2.1.3 to determine conformance with 3.3.1.1.

4.6.1.3 The hydrometer shall be placed in boiling (212°F.) fresh water for 1 hour The test specified in 4 6.1.2 shall be repeated. There shall be no difference in readings obtained before boiling and after boiling.

4.6.2 Time lag. - The time lag of the thermometer shall be determined by the following procedure: Two agitated liquid baths at temperatures near the upper and lower limits of the thermometer shall be used. The bulb shall be immersed up to the top of the guard in the hot bath for 10 minutes, and then in the cold bath for 2 minutes. The bulb shall be quickly transferred to the hot bath and immersed to the top of the guard. Timing shall be started when the reading of the thermometer passes an arbitrary point at least 25°F. above the temperature of the cold bath. Timing shall be stopped when the reading of the thermometer passes a point at 63 percent of the interval between the reading at the start of timing and the temperature of the hot bath The timing interval shall be at least 25 percent of the range of the thermometer. The time lag tests shall be made with a water bath

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (see 6.1). -

5.1.1 Level A. - Salinometer equipment shall be processed as follows:

5.1.1.1 Hydrometer. - Hydrometers shall be individually preserved and packaged in accordance with method III of Specification MIL-P-116. Each hydrometer shall be wrapped in neutral cellulosic cushioning material and packaged in a snug fitting set-up paperboard box conforming to Specification PPP-B-676. Box closure shall be as specified in the appendix to the box specification. Telescoping style boxes shall be closed with paper gummed tape and applied in such a manner so as to completely seal the closure.

5.1.1.2 Thermometers. - Thermometers shall be individually preserved and packaged in accordance with method III of Specification MIL-P-116 Thermometers shall be individually wrapped in neutral tissue paper of 10 pounds basis weight (480 sheets - 24 by 36 inches) or neutral cellulosic cushioning material and packaged in a snug fitting set-up paperboard box conforming to Specification PPP-B-676. Box closure shall be as specified in the appendix to the box specification Telescoping style boxes shall be closed with paper gummed tape and applied in such a manner so as to completely seal the closure

5.1 1.3 Test pot - Test pots shall be individually preserved and packaged in accordance with method III of Specification MIL-P-116.

5.1 2 Level C. - Salinometer equipment shall be preserved and packaged in such a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use This may conform to the supplier commercial practice, when such meets the requirements for this level

5 2 Packing (see 6.1) -

5 2 1 Level A - Salinometer equipment packaged as specified, shall be packed in wood cleated fiberboard, nailed wood, fiber, wirebound wood, wood cleated veneer paper overlaid or wood cleated plywood boxes conforming to Specification PPP-B-591 (overseas type), PPP-B-621 (class 2), PPP-B-636 (class 2), PPP-B-585 (class 3), MIL-B-10377 (overseas type) or PPP-B-601 (overseas type) respectively at the option of the contractor Shipping containers shall have caseliners conforming to Specification MIL-L-10547 Caseliners shall be closed and sealed in accordance with the appendix to Specification MIL-L-10547 Caseliners for boxes conforming to Specification PPP-B-636 may be omitted provided all corners and edge seams and manufacturer's joints are sealed with a minimum 1-1/2 inch wide tape conforming to Specification PPP-T-60 Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto, except fiber boxes shall be banded with tape conforming to type IV of Specification PPP-T-97 and applied in accordance with the appendix thereto The gross weight of wood or wood cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification

5 2.2 Level B. - Salinometer equipment, packaged as specified, shall be packed in wood-cleated

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fiberboard, nailed wood, wirebound wood, wood cleated plywood, wood cleated veneer paper overlaid or fiber boxes conforming to Specification PPP-B-591 (domestic type), PPP-B-621 (class 1), PPP-B-585 (class 1), PPP-B-601 (domestic type), MIL-B-10377 (domestic type) or PPP-B-636 (class 2), respectively at the option of the contractor Box closure shall be as specified in the applicable box specification or appendix thereto The gross weight of wood or wood cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification

5 2 3 Level C. - Salinometer equipment, packaged as specified, shall be packed in containers which will ensure acceptance by common carrier and safe delivery to destination. Shipping containers shall comply with the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation

5 3 Marking - In addition to any special marking required by the contract order, interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129

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6. NOTES

6 1 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of individual items required (see 3 2)
(c) Selection of applicable levels of preservations, packaging, and packing (see 5.1 and 5 2).

Notice - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto

Preparing activity.
Navy - Ships
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